Entertainment Computing 54 (2025) 100942

Contents lists available at ScienceDirect

Entertainment Computing

journal homepage: www.elsevier.com/locate/entcom

Using gamification to enhance learning: A college course case study

Chih-Chien Wang [a], Shu-Chen Chang [b,*], Yu Han Yu [a]

[a] Graduate Institute of Information Management, National Taipei University, Taiwan (R.O.C.)
[b] Department of Business Administration, National Taipei University of Business, No.321, Sec. 1, Jinan Rd., Zhongzheng District, Taipei City 100, Taiwan (R.O.C.)

ARTICLE INFO

Keywords:
Gamification
Gamified design
Learning engagement
Learning effect
Classcraft

ABSTRACT

Gamification means using game design elements, such as rules, structure, voluntary participation, uncertain outcomes, conflict, decision-making, and outcomes, in non-game contexts. The study compares the differences in learning outcomes and engagement between gamified and traditional instruction methods in university classrooms. Two groups of university students were taught by the same teacher, using the same materials and exams but with different teaching designs. The experimental group used the gamified software Classcraft for assignments, which involved points, badges, leaderboards, and social and entertaining game features, while the control group used traditional lectures and assignments. The results showed a significant improvement in the exam scores of the experimental group of gamified instruction compared to the control group of traditional lecture instruction.

Additionally, the study found that gamified instruction has several positive effects, such as reducing learning stress, increasing course enjoyment, and enhancing both in-class engagement and after-school learning motivation. Gamified instruction enhances learning engagement, which positively influences academic performance and increases the range of improvement. This study provides empirical evidence for using gamified teaching in university education and discusses the design principles and practical applications of gamified teaching. The study concludes that gamification can be used in university classrooms to improve student engagement and learning performance.

1. Introduction

Improving students' learning performance is a primary goal of educators. However, teachers often need help engaging students who may lack motivation or interest in the subject and tend to be apathetic. In traditional teaching models, teachers deliver course content from the front of the classroom while students sit and listen. When teachers occasionally pose questions to the students, the responses are typically unenthusiastic.

The game industry has a long history and has profoundly influenced society in various aspects. Games often involve imitation, competition, or chance. Children naturally engage in games without explicit instruction, and through this play, they unconsciously imitate and internalize social and cultural norms. Research has explored the elements that constitute games, which have been applied in non-game contexts [1,2], a process referred to as gamification. Gamification primarily focuses on creating enjoyable experiences by incorporating engaging storylines, challenges, enticing rewards, and a sense of achievement through recognition or rankings. Gamification has been widely employed in education (to facilitate learning), health care (to encourage physical activity), and employee management (to promote proactive organizational behavior), among other contexts [3].

Numerous educational studies have explored the integration of teaching and games to promote student learning engagement and motivation [4–7]. Literature has demonstrated that gamification positively influences learning and improves learning outcomes [8]. Numerous studies have demonstrated that gamification provides various instructional benefits, including real-time feedback, fostering healthy competition, enabling self-paced learning, and encouraging teamwork. [9–13].

In education settings, gamified teaching designs can vary considerably. Many scholars have conducted experiments and analyses involving different gamified instructional methods [6,7,14]. Different gamified teaching methods may yield different learning outcomes. Training needs, design principles, and evaluation methods also constrain the effectiveness of gamified training. Game elements should not be added

* Corresponding author.
E-mail address: Susan.c@ntub.edu.tw (S.-C. Chang).

https://doi.org/10.1016/j.entcom.2025.100942
Received 7 August 2023; Received in revised form 12 September 2024; Accepted 19 March 2025
Available online 22 March 2025
1875-9521/

arbitrarily; instead, they should align with the goals and processes of training [15].

While most research on gamification in education has focused on high school students, gamified instruction in university classrooms remains relatively uncommon [16,17]. College students often come from different backgrounds and departments with varying learning purposes. The training needs of college students are diverse, and the principles of instructional design must be considered in multiple ways. Additionally, teachers might sacrifice time in implementing gamified elements, potentially causing delays in course progress. These are some of the practical challenges that may arise in implementing gamified instruction in university classrooms.

Therefore, this study incorporated gamified instruction into a university course to explore its influence on student learning outcomes. Critics argue that employing point systems, badges, and leaderboards for gamification represents superficial efforts to motivate learning. So, this study not only adopted the common gamified elements such as game points, competition, and leaderboards [18]but also provided the addition of a story, a character's virtual appearance, reinforcements, and social interaction. This study investigates the effects of gamified instruction on learning performance and engagement in a university course context.

The main objective of this study is to examine the impact of gamified instruction on learning performance in a college course. Learning performance is measured by improving exam scores from pretest to posttest. The secondary objective is to explore the relationship between learning engagement and performance. A questionnaire and the scores of gamified tasks and quizzes measure learning engagement. The participants are undergraduate students in the business field assigned to either an experimental or a control group.

This study adopted an experimental design to achieve the objectives and used the gamification software Classcraft to implement gamified instruction in a university course. The gamified software is used after school rather than during the course hours. During weekly class sessions, the teacher reviewed the gamification status and lectured on the teaching content following the syllabus. The research questions of this study are as follows:

RQ1: Does gamified course design enhance learning performance in a university course?

RQ2: Does learning engagement resulting from gamified course design positively influence learning performance?

RQ3: Do the experience points from gamified learning activities positively influence learning performance?

2. Literature

2.1. Games and gamification

Games were defined in early research as voluntary activities constrained by rules and involving conflict and unequal outcomes among equal participants [19]. Juul et al. [20] reported that all games have six main characteristics: rules, variable and quantifiable outcomes, valuable outcomes, player effort, player investment, and negotiable consequences. Salen et al. [21] explained that games are systems in which players engage in artificial conflicts defined by rules, with the conflicts leading to quantifiable outcomes. A game is a combination of necessary conditions that are individually insufficient to represent a game's characteristics but, when combined, constitute a game [22,23].

Based on the definitions above, we can consider a game as a set of activities that combine rules, structure, voluntary participation, uncertain outcomes, conflict, decision-making, and outcomes. Beyond games, gamification is not quite the same with games.

Based on the early interpretations of gamification, it involved using game design elements in non-game contexts [15,22,24]. Thus, we define gamification as "using game design elements of rules, structure, voluntary participation, uncertain outcomes, conflict, decision-making, and outcomes on non-game contexts.".

Several factors have influenced the emergence of gamification, including increased availability of affordable, relevant technology, personal data tracking, success in its application, and the popularity of gaming media [25]. Literature advocates that the effects of game elements could increase motivation and involvement in an educational context in various ways [26]. The effects of gamification are inconsistent but depend on different game elements, learning activities, and student characteristics. Games are enjoyable activities, but gamification teaching is an education strategy. Players engage in games for fun, while students participate in gamification for learning. While gamification fits class objectives, games fit player preferences. The game has unchallengeable rules and predictable results [27], but the gamification is flexible.

2.2. Gamified learning and game-based learning

Werbach and Hunter [28] described gamification as using game elements and game design techniques in non-game contexts to motivate individuals to accomplish specific tasks. Gamification enhances participants' gaming experiences by increasing value creation for users [29]. Participants determine whether something constitutes a game or involves gamification [3].

The main difference between game-based learning and gamified learning lies in how they incorporate game elements into the learning process. Game-based learning involves using games to teach specific skills or achieve specific learning outcomes [30]. In game-based learning, the game is part of the instruction, and learners can learn and practice knowledge points through the game. For example, digital dialogue games can train students and stimulate their motivation for learning argumentation [31–33]. Educational games are a common form of Game-based Learning, which turns instructional content into a game interface. In education, educational games are generally believed to enhance the retention of knowledge and skills acquired through learning and to promote learning engagement, motivation, and interest in the learning process [34–36].

Gamified learning, on the other hand, applies game principles to the learning process. It increases learner engagement and motivation by introducing game mechanics (such as points, levels, and rewards) [26]. Gamified elements enhance the attractiveness and effectiveness of learning, even though the learning process is not a game. The effectiveness of gamified learning depends on selecting and applying game elements, with different elements influencing different learning outcomes [15].

In summary, both game-based learning and gamified learning leverage the appeal of games to enhance learning effectiveness, but they differ in their implementation methods and objectives.

2.3. Gamified features

Wang et al. [24] indicated six fundamental gamified design principles: integration with training goals, rapid feedback, team competition, clear rules, goal-oriented challenges, and freedom to fail. Ahmad et al. [26] showed a gamification framework that identifies five gamification constructs: Goal orientation, achievements, reinforcements, competition, and fun orientation. Dicheva et al. [37] pointed out some common gamification design principles: Goals and challenges, personalization, rapid feedback, visible feedback, freedom of choice, freedom to fail, and social engagement. These features of gamification are categorized and explained as follows:

2.3.1. Reinforcement

In the psychological behavior learning model, learning is characterized by negative and positive reinforcements [38]. Among the gamified elements, badges and points are for positive reinforcement.

Within the framework of operant conditioning theory, points and badges serve as reinforcements, encouraging students to meet instructional expectations and accomplish specific tasks or objectives [24,39]. The points and badges also promote gamified learning by providing rapid feedback and competition [18,26].

A leaderboard was also a common element of game mechanics. Ranks and leaderboards are employed for social comparison [40]. Self-regulation theories can be employed to elucidate the application of leaderboards in motivating students to undertake more ambitious objectives[41]. Students may also be motivated by being placed higher on the leaderboard than others. The motivation is intrinsic self-regulatory.

The overall positive results related to the gamification elements of points, badges, and leaderboards indicate their potential for practical integration into training designs to enhance motivation and learning performance [15]. Reinforcement is a common result of these gamification elements.

2.3.2. Freedom of choice and freedom to fail

In the game context, users have the freedom to choose and can afford the negative outcome of failure. The foundation of university courses is diverse students' visions; if students are afraid to make mistakes, they might step back and discourage their learning motivations. If the incorrect behavior or attitudes did not cause negative consequences, students might get more involved in class practice after school. The freedom of choice and freedom to fail features of gamification enhance the effectiveness of the training program, especially if the emphasis is on addressing students' incorrect behavior or attitudes[42]. In the real world, incorrect or inappropriate choices may bring negative outcomes. Gamified designs provide users a chance to know the consequences of inappropriate choices, which is a useful and meaningful feature for educational purposes.

2.3.3. Fun orientation and social orientation

The fun and social orientation are also gamification features. Gamified with social interaction features may raise students' engagement in a learning context. Social features might tighten students together in class to face challenges[43]. Additionally, game features such as virtual goods, virtual pets, and virtual characters' appearances are fun-orientation features that might strengthen the enjoyment and engagement in gamified learning.

2.4. Engagement

The term 'engagement' is widely used to refer to various forms of participation in activities[44]. In student learning, engagement encompasses elements such as attendance, academic participation, and interactions between teachers and classmates. Gamification in education helps increase students' motivation to achieve specific goals and complete tasks[45,46], and therefore, gamification enhances student engagement with course content and increases teacher involvement and interaction with students [1].

Research has also demonstrated that using technology or social media as instructional tools can enhance student learning [14,47–50]. Such technology offers channels for interaction, such as discussion boards and instant messaging, which can facilitate interaction between students and teachers as well as among students. In addition, it provides tools that can be used for academic activities, such as note-taking. Thus, technology use can promote learning engagement and self-directed learning [51–56]. Furthermore, students can collaborate with other students in academic activities by using online messaging, which enhances learning engagement [57]. Technology can improve interpersonal relationships and learning engagement by deepening the connections among students, teachers, and instructional content [58].

Literature indicates that student learning outcomes are influenced by various factors, including learning engagement, which correlates with academic achievement [59–62]. Student learning engagement in a course can predict learning outcomes and personal development [59]. The more frequently and intensely students engage with a subject or practice a skill, the more likely they will acquire relevant knowledge and proficiency. The more students practice reading, analyzing, problem-solving and receiving feedback, the more skilled they become [63]. Higher levels of learning engagement led to superior academic performance [64–66]. Furthermore, learning engagement is the foundation for skill and character development. Student learning engagement in higher education facilitates the development of study strategies, habits, and mindsets, strengthening students' capacity for continual learning and personal growth[67].

Based on the discussion mentioned above, the current study proposed the following hypotheses:

H1: Gamified instruction can enhance learning outcomes.
H2: Learning engagement positively influences learning outcomes.

3. Method

3.1. Procedure

This study adopted an experimental design. The experiment involved two groups of university students taught by the same teacher using the same course content. An experimental group, and a control group. The experimental group received gamified instruction provided using the gamification software Classcraft. The control group received traditional classroom lectures without gamification. Both groups covered the same class content according to the same syllabus; the gamified software Classcraft was the only difference, applied to the experimental group to incorporate gamification elements.

In the pre-course phase, both groups took an initial assessment of course knowledge in the first week to measure the expertise and subject-specific information students possessed before attending the class. After the testing, all students complete a learning engagement scale to assess the engagement levels in the course and an open-ended questionnaire to give feedback on their expectations about the ongoing class.

During the course, the experimental groups adopted the gamified course design (see 3.2 section), and the control group took traditional course lectures. All review and self-learning quiz materials were sent to both groups. In the experimental group, the quiz materials were designed as game levels for gamification purposes, and game points were gained as students progressed through these levels. In control groups, the quiz materials were assigned in Google Forms. All students took a final examination and another open-ended feedback questionnaire at the post-course stage. The improvements in learning outcomes were assessed by comparing the pretest and posttest scores. This stage also included in-depth interviews with four students from the control group and six from the experimental group. The study procedure is shown in Fig. 1.

3.2. The tool: Classcraft

Classcraft [68]is a commercially available gamification platform for educational settings. As a professional educational technology product, Classcraft offers comprehensive gamification features that can be seamlessly integrated into various courses. Considering Classcraft's robust functionality and maturity, this study specifically subscribed and applied it to the research design.

Classcraft is a gamified role-playing learning environment where users can select from various characters with varying skills, such as warriors, mages, and healers. Each character has gold points, lives, and energy. The teacher created the characters' abilities and gave them to various characters. "I can remit a delay record in class, such as homework delay or arriving late to class," is an example of an ability. To unlock more abilities, students must accumulate more points.

Classcraft offers numerous story levels and a few war stories. The

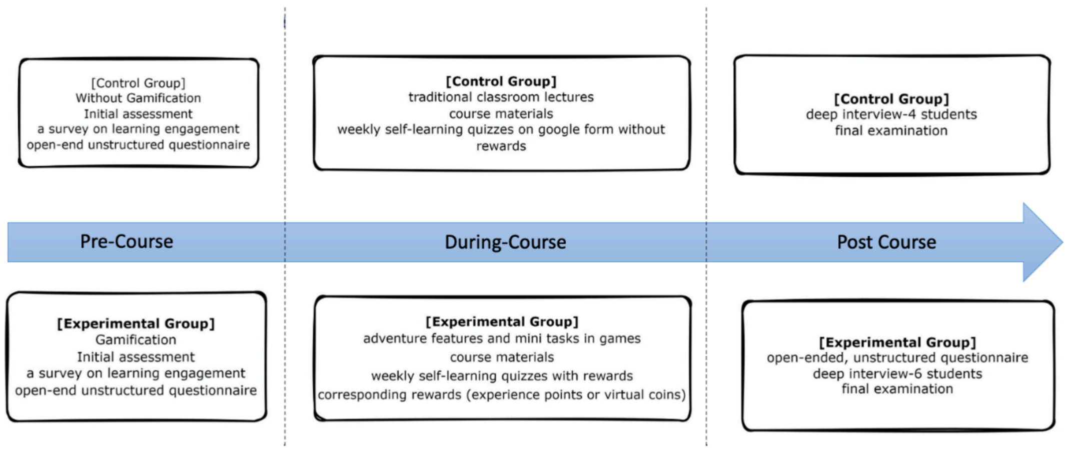

Fig. 1. Study Procedure.

content and level names are up to the teachers. Classcraft offers extra features that encourage student participation, such as quiz content and course knowledge. The ability for students to alter their appearance, own a pet, play with the wheel of destiny, discover treasures, and more makes the class more engaging.

3.3. Gamified course design

Classcraft has a visually appealing interface and comprehensive features. It can be accessed on various devices, including computers, tablets, and smartphones. It enhances interaction between teachers and students and promotes student course engagement. In the experimental group, the adventure feature of Classcraft software was utilized to present gamified tasks. These tasks included participating in weekly self-learning quizzes, reading course materials, and completing assignments. Each task provided rapid feedback and corresponding rewards, such as experience points or virtual coins. Following each weekly class, the teacher granted the experimental group access to the gamified task for one week and encouraged students to self-learn during their free time. Students could attempt each task unlimited times, alleviating any fear of failure. The exploration feature of Classcraft was used to provide supplementary materials for the course that could be downloaded. This study adopted Classcraft software due to its interactive features, facilitating teacher-student interaction and enhancing students' engagement with the gamified tasks and the course.

Specific gamified features, such as character appearances and the pet system, were also incorporated to enhance enjoyment. For example, the 'Wall of Glory' feature in Classcraft allows peers to commend each other, with these accolades visible to everyone. The teacher publicly acknowledges these praises in the classroom. The details of gamified elements and constructs are shown in Fig. 2.

3.4. Sample

This study recruited participants from undergraduate students in the Department of Advertising, specializing in commercial design, advertising strategies, and digital advertising. The study was conducted in a statistics course to enhance students' understanding of market data, consumer behavior analysis, and data-driven marketing strategy formulation. Classcraft helps students review their statistical knowledge after school.

Initially, the experimental group comprised 54 students, and the control group was 69 students. After excluding incomplete data (e.g., missing questionnaires, absence from exams, course dropouts), the final valid sample consisted of 43 students in the experimental group and 50 in the control group.

All participants were sophomore undergraduate students. The experimental group comprised 37 females and six males, while the control group had 39 females and 11 males. A chi-square test revealed no significant difference in gender distribution between groups ($p = 0.317$). Chi-square tests were conducted to ensure no significant differences in gender and age distributions between the experimental and control groups, thus confirming the comparability of the samples. The detail is shown in Table 1.

Among the 43 students in the experimental group, 5 were 19 years old, 29 were 20 years old, 4 were 21 years old, 3 were 22 years old, 1 was 23 years old, and 1 was $\geq$ 24 years old. Among the 50 students in the control group, 11 were 19 years old, 28 were 20 years old, 3 were 21 years old, 5 were 22 years old, 2 were 23 years old, and 1 was $\geq$ 24 years old. A chi-square test revealed no significant difference in the age ratio

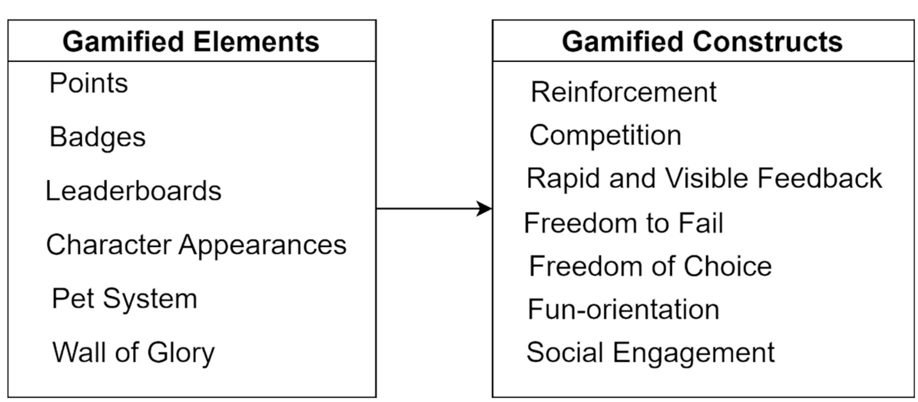

Fig. 2. Gamified Elements and Constructs.

Table 1
The details of gender analyses between the experimental group and control group.

Experimental group	Participants	Control group	Participants
Male	6	**Male**	11
Female	37	**Female**	39
All	43	**All**	50

chi-square test in the gender ratio between the experimental and control groups

	Value	**df**	**p-value**
X^2	1.002	1	0.317
N	93		

between the experimental and control groups ($p = 0.741$). The detail is shown in Table 2.

The final sample was representative of the target population, with no significant differences in critical demographics between the experimental and control groups.

3.5. Ethic

This study was exempted from Institutional Review Board approval as per the Food & Drug Administration, Ministry of Health and Welfare, Taiwan, Order No. 1010265075, as this study belongs to a study of educational assessment or testing, teaching skills or effectiveness evaluation conducted in a general teaching environment. All participants in the study were not minors, prisoners, aborigines, pregnant women, physical illness, or mental illness, and were not improperly intimidated or unable to make decisions of their own free will.

This study announced the study design and purpose to students in the first week of the course. Students were informed that they were involved in a study. They were informed that exam scores, quiz scores, Classcraft usage, open-ended questionnaires, and deep interviewing content were used for academic purposes. They hold the right not to join the study, and the join or not would not influence their course score. The final exam is the final evaluation of the course. The data was non-anonymous but would not be public.

4. Measurement

4.1. Learning outcomes

This study assessed learning outcomes when implementing gamified instruction by evaluating students' academic performance and engagement. Students completed identical exams at the beginning and end of the semester. Each exam consisted of 100 multiple-choice questions, with each question worth 1 point. The maximum score was 100, and the minimum score was 0, with no negative marking for incorrect answers. While the core knowledge, concepts, calculation methods, and number of questions were identical in both exams, the final exam featured modified descriptions, nouns, units, and values. Additionally, the order of questions and answer choices was randomized to prevent direct memorization and ensure valid measurement of learning outcomes. Learning outcomes were analyzed by calculating students' final and initial exam scores. This approach allowed for a direct comparison of student performance before and after the implementation of gamified instruction, providing a quantitative measure of learning improvement.

Table 2
The details of age analyses between the experimental group and control group.

	Age						
	19	**20**	**21**	**22**	**23**	**Over 24**	**Total**
Experimental group	5	29	4	3	1	1	46
Control group	11	28	3	5	2	1	50
Total	16	57	7	8	3	2	93

chi-square test in the age ratio between the experimental and control groups

	Value	**df**	**p-value**
X^2	2.732	5	0.741
N	93		

4.2. Engagement

This study utilized the Student Course Engagement Questionnaire (SCEQ) developed by Handelsman et al. [69] to measure learning engagement. The students responded to the questionnaire items on a 5-point Likert scale, ranging from 1(strongly disagree) to 5 (strongly agree).

Following each week's course lecture, all students in both groups were provided with self-learning materials to reinforce course content. Quizzes were designed to accompany each piece of self-learning material, assessing students' understanding and engagement with the content. For the experimental group, engagement was measured using quiz scores and the experience points obtained through completing gamified tasks in Classcraft. While self-study materials and quizzes were also provided to the control group, their quiz scores were not counted towards engagement measurement. Classcraft experience points were not available for the control group. This difference in measurement approach was implemented because quantifiable outcomes, such as game points, are integral elements of gamification. Including these measures for the control group would have introduced inconsistencies in the engagement assessment between the two groups.

4.3. Reliability and validity

The Cronbach's α value for the measure of learning engagement was 0.946, which exceeds the acceptable threshold of 0.7, indicating high scale reliability. The composite reliability value for the measure of learning engagement was acceptable at 0.964, exceeding the threshold of 0.7. This study calculated the average variance extracted (AVE) to assess convergent validity and obtained a value of 0.549, which exceeds the acceptable threshold of 0.5. The reliability, composite reliability, and convergent validity of the learning engagement scale all met the recommended levels [70], indicating that the study's measure of learning engagement had high reliability and validity. Details are shown in Table 3 **reliability and validity**.

A Likert-type scale was used to measure learning engagement. Learning performance was measured as exam score improvement. Learning engagement and learning performance were not measured using the same approach. Thus, it was unnecessary to consider the discriminant validity between the learning engagement and performance measures.

5. Result

5.1. Gamification and learning performance

This study found that gamified instruction significantly improved learning performance compared to traditional instruction. The experimental group (gamified instruction) showed an average improvement of 28.12 points (SD = 18.76) from the initial to the final exam, while the control group (traditional instruction) improved by 18.1 points (SD = 12.46), as shown in

Table 3
Reliability and validity.

	learning engagement	Acceptable range
Cronbach's α	0.946	> 0.7
composite reliability	0.964	> 0.7
average variance extracted (AVE)	0.549	>0.5

Table 4. An independent samples *t*-test revealed a significant difference in exam scores and the improvement between the experimental and control groups ($t = 3.070$, $p = 0.003$), as shown in Table 5.

In Table 5, this study also investigated the association between the initial and final exam scores and learning performance (score improvement). The initial exam scores did not differ significantly between the experimental and control groups ($t = -0.752$, $p = 0.454$). However, the final exam ($t = 2.613$, $p = 0.01$) and learning performance (score improvement; $t = 3.070$, $p = 0.003$) scores did. In Table 6, the correlation coefficient for the initial scores and score improvement was 0.097, indicating no significant correlation ($p = 0.536$).

In the control group of Table 6, a significant negative correlation between the initial scores and score improvement was noted, with a correlation coefficient of -0.335 ($p = 0.017$) being obtained. This finding indicates that students with more excellent prior knowledge (higher initial exam scores) exhibited less learning performance (lower score improvement) under traditional teaching conditions. Thus, the traditional teaching model benefited students with less prior knowledge.

The results of the analysis support H1: Gamified instruction can enhance learning outcomes.

5.2. Engagement and learning performance

This study also conducted a correlation analysis of the experimental group's quiz scores, Classcraft experience scores, and exam score improvement. Each student's quiz score was the cumulative total of their highest scores on each quiz completed in Classcraft. Students were encouraged to retake quizzes multiple times to improve their understanding and scores.

Classcraft experience scores were calculated at the semester's end, reflecting students' cumulative engagement in the gamified learning environment. These scores were derived from various activities within the Classcrift system, including classroom interactions, quiz completions, and participation in gamified tasks. An interesting feature was the pet nurturing system, which, enhanced overall engagement while not directly contributing to the learning outcomes. Students could nurture their virtual pets using coins earned from quizzes, creating an additional incentive for quiz participation. As students interacted with course materials and completed tasks, their experience points steadily accumulated. The final accumulation of these points provided a quantitative measure of each student's engagement level in the gamified learning process throughout the semester.

Analysis revealed in Table 7 a correlation coefficient of 0.698 ($p < 0.001$) for the quiz scores and Classcraft experience scores. This result is reasonable because the quiz scores contributed to the total experience scores. This study obtained a correlation coefficient of 0.926($p < 0.001$) for the weekly quiz scores and score improvement, which indicates that the students with higher scores on the weekly quizzes exhibited more remarkable improvement in their overall scores, suggesting that the students who conscientiously completed weekly quizzes achieved superior learning outcomes. Additionally, the correlation coefficient for Classcraft experience scores and score improvement was 0.62 ($p < 0.001$), indicating that the more actively the students participated in the gamified learning, the better their learning outcomes were.

The statistical analysis results support hypothesis H2: Learning engagement positively impacts learning outcomes.

In terms of Table 8 result, this study analyzed covariance (ANCOVA) to investigate the influence of learning engagement and gamified instruction on learning performance (exam score improvement). The analysis revealed a p-value for the factor "groups" (experimental/control) of 0.004, indicating that the independent variable "experimental group" had a significant explanatory power on the dependent variable "learning performance" after the covariate "learning engagement" was controlled for. This finding further validated H1.

The covariate "learning engagement" had a p-value of 0.004, indicating that it significantly explained learning outcomes the results of this statistical analysis support H2.

5.3. Student feedback

This study employed multiple feedback methods throughout the course to gain insight into students' experiences and perceptions of both traditional classroom and gamified instruction. At the beginning of the semester, an open-ended, unstructured questionnaire was distributed to all students to gather their initial expectations and opinions about the course. Upon completion of the semester, students in the experimental group completed another open-ended, unstructured questionnaire to assess whether the gamified instruction met the students' expectations.

Table 5
Independent t *t*-test results for Initial Exam Scores, Final Exam Scores, and Score Improvement in Experimental and Control Groups.

	t	df	p	Cohen's d
Initial exam score	−0.752	91	0.454	−0.156
Final exam score	2.613	91	0.01	0.544
Score improvement	3.070	91	0.003	0.639

Table 6
Correlation Coefficients for Initial Exam Scores, Final Exam Scores, and Score Improvement.

		Initial exam score	Final exam score
Experimental group	Final exam score	$0.416(p = 0.006)^{**}$ Fisher's Zr = 0.443	—
	Score improvement	$0.097(p = 0.536)$ Fisher's Zr = 0.097	$0.945(p < 0.001)^{***}$ Fisher's Zr = 1.7828
Control group	Final exam score	$0.210(p = 0.143)$ Fisher's Zr = 0.213	—
	Score improvement	$-0.335(p = 0.017)^{*}$ Fisher's Zr = -0.348	$0.851(p < 0.001)^{***}$ Fisher's Zr = 1.260

* $p < 0.05$,
** $p < 0.01$,
*** $p < 0.001$.

Table 7
Correlation coefficients for Quiz Scores, Classcraft Experience Points, and Exam Score Improvement.

	Quiz scores	Exam score improvement
Exam score improvement	$0.926(p < 0.001)^{***}$ Fisher's Zr = 1.630	—
Classcraft experience points	0.698 ($p < 0.001)^{***}$ Fisher's Zr = 0.8634	0.62 ($p < 0.001)^{***}$ Fisher's Zr = 0.725

*** $p < 0.001$.

Table 4
Initial Exam Scores, Final Exam Scores, and Score Improvement.

	Initial exam score		Final exam score		Score improvement	
	Experimental group	Control group	Experimental group	Control group	Experimental group	Control group
Sample size	43	50	43	50	43	50
Mean	35.79	36.84	63.91	54.94	28.12	18.10
Standard deviation	6.72	6.70	20.53	12.01	18.76	12.46

Table 8
Analysis of Covariance Results (ANCOVA).

Source	Sum of squares	Degree of freedom	Mean square	F	*p*	*Estimates of Effect Size* η^2
Group	1717.37	1	1717.371	8.663	0.004	0.102
Engagement	1740.11	1	1740.11	8.777	0.004	0.043
Error	17842.76	90	198.253			

Dependent variable: Learning performance.

At the end of the semester, we conducted in-depth interviews with ten students, six from the experimental group and four from the control group, to further explore the suitability of gamified and traditional instruction for different students. The interviews were conducted via online conference meetings on Google Meet. The interviews investigated the association between the student's academic performance and gamified instruction. Interviewees were selected based on their performance, including those with high or low scores on the initial and final exams, to ensure diverse perspectives. Each student was asked about their study habits, learning status, final exam preparation, and thoughts and experiences regarding the teaching mode used during the semester. We summarize the key findings from student feedback and categorize them into the pros and cons of both gamified and traditional instruction, as shown in Table 9.

Most students had actively used Classcraft and engaged with the gamified instruction. Whenever a new level in Classcraft was unlocked, the students enthusiastically completed new quizzes. The students reported that completing exams in a game format reduced their study pressure and allowed them to relax when reviewing the course content. Additionally, the lack of a limit on the number of quiz attempts enabled them to take the exams repeatedly and improve their learning outcomes.

According to the student feedback in Table 9, the gamified instruction enhanced learning motivation by making the course more enjoyable. Gamified instruction enabled course content visualization, helped students understand the learning material, and increased their sense of achievement, thus reducing learning pressure. However, the selected gamified instructional software, Classcraft, occasionally exhibited system operation problems. The interview results revealed that most students disliked traditional classroom teaching, considering it monotonous and ineffective. Nevertheless, traditional teaching methods remain viable for students who do not enjoy playing computer games.

Table 9
Interview Summary from Student Feedback.

Categories	Details
Pros of gamified instruction	• Interactive discussions enabled by gamification in the classroom helped students focus and deepened their learning. • Gamified instruction enhanced student learning motivation because the games were enjoyable. • Gamified instruction added enjoyment to the course and enhanced learning motivation. A sense of competition was created through gamified instruction, which boosted student learning motivation. • Gamified instruction can assist students in the understanding of the learning content. • Gamified instruction increases students' sense of achievement in learning. • Gamified instruction reduces learning stress.
Cons of gamified instruction	• The game system occasionally did not operate smoothly, especially on older mobile phones with lower technical specifications. • Navigating the game was time-consuming. • Gamified instruction reduces students' learning efficiency with a high level of prior knowledge. • The gamified instruction was unsuitable for students who did not enjoy playing computer games.
Pros of traditional classroom instruction	• Traditional instruction did not require students to spend time using gamification software. • For highly motivated students, traditional classroom lectures enable high-efficiency learning.
Cons of traditional classroom instruction	• Traditional instruction can be boring for students. Traditional teaching methods often result in dull and uninteresting courses, making concentrating easier and leading to better learning outcomes. • Traditional instruction leads to low learning efficiency in students with low learning motivation.

6. Discussion and Conclusions

Gamified instruction outperformed traditional lecture instruction under the same conditions (i.e., same teacher, course content, and exam content), demonstrating that gamified instruction can enhance learning effectiveness. Furthermore, learning engagement has a positive influence on learning outcomes. Course design can enhance learning engagement, thereby improving learning effectiveness.

Unlike previous studies that primarily use points, badges, and leaderboards, this research incorporates additional elements such as story context, virtual character appearances, pet systems, and social interactions. These elements make the gamified teaching more engaging and appealing. Points, badges, and leaderboards were commonly used in the classroom to stimulate engagement; however, they can lead to a lag in the teaching schedule if overused during class time. A key distinction of this study is incorporating most gamification elements into after-class activities. This approach stimulates students' intrinsic motivation to actively participate in a gamified learning environment outside class time while minimizing disruption in class teaching schedules.

In this study, gamification rules are predefined, and students carry them out after class. The teacher's primary role in this gamified teaching design is to review students' weekly performance in the classroom, effectively bridging the motivational effects of gamification with in-class activities. This study precisely measures the effectiveness of gamification by comparing the final exam scores. The gamification results, such as points, badges, leaderboards, character appearances, and pet development, will not be given extra points in learning outcomes. Therefore, the gamified elements of this study stimulate students' intrinsic motivation in various ways and do not stimulate students' extrinsic motivation.

The most significant value of this study lies in the fact that gamification elements are balanced with the progress of the course. Moreover, previous gamification research often mixed gamification with score rewards, using extrinsic motivation to stimulate students. This study does not use score incentives at all and does not utilize extrinsic motivation, but it can still improve learning outcomes. The results indicate that the enhancement of learning motivation generated by gamification can motivate students without relying on the extrinsic motivation of extra points. Reducing the dependence on extrinsic motivation can make learning an autonomous activity conducive to cultivating students' self-learning habits. This study's qualitative data analysis revealed that students dislike traditional lecture teaching because it is monotonous, which leads to lower student attention and ineffective learning, resulting in lower learning performance. Many students preferred engaging in interactive discussions facilitated by games, and enjoyment of the games motivated the students to participate actively in the course. Games also created a relaxed atmosphere, making the students more willing to engage with challenging learning materials. Playing games also improves student competitiveness, leading to more active learning and improved learning outcomes. Additionally, games visualize learning content, learning progress, and achievements, making understanding key concepts, tracking progress, and gaining a sense of accomplishment easier for students, which enhances their learning effectiveness and engagement.

However, in the traditional teaching context, the improvement range is higher for students with less prior knowledge. This finding is shown in Table 6 and indicates that the structured teaching method is suitable for students who have lower prior knowledge. Moreover, regarding the feedback in Table 9, the gamified instruction reduced the high level of prior knowledge students' learning efficiency. The gamified instruction was also unsuitable for students who did not enjoy playing computer games.

While this study shows the positive effects of gamified instruction in a single course over one semester, future research should explore its effectiveness across multiple courses and over more extended periods.

This research provides empirical evidence for the effectiveness of gamified instruction in university settings, offering a model for balancing gamification elements with course progress and fostering intrinsic motivation in students.

7. Theoretical Implications

This study contributes to the growing body of literature on gamification in higher education, addressing a significant gap in research on university-level applications. Unlike previous studies that primarily focused on primary or secondary education [47,52,59], this research provides empirical evidence for the effectiveness of gamified instruction in a university setting. Despite the previous implementation of Classcraft in university courses, this study thoroughly analyzes its impact on learning outcomes and engagement, backed by objective experimental data. The findings of this study advance our understanding of the relationship between gamified instruction, learning engagement, and learning outcomes in higher education contexts.

8. Research limitations

This study assessed learning engagement through a self-reported questionnaire. While this approach provided valuable insights, it may lack objectivity. For a more comprehensive analysis of learning engagement, future research should incorporate additional objective data, such as attendance records, punctuality, frequency of quiz attempts, quiz scores, and student-instructor interactions. Furthermore, while Classcraft provided relevant learning engagement data for the experimental group, comparable objective data for the control group was limited, potentially affecting the comparability between the two groups. Future studies should address this limitation by ensuring equivalent data collection methods for both experimental and control groups, allowing for a more comprehensive analysis of learning engagement.

While this study demonstrated the positive effects of gamified instruction on learning outcomes over a single semester in one course, its generalizability to multiple courses or more extended time frames remains uncertain. Previous longitudinal research on game-based instruction has shown promising results over two years [2]. Further, longitudinal studies are needed to confirm the long-term impact of gamified instruction and learning engagement on learning outcomes and explore potential differences between gamification's short-term and long-term effects in education.

9. Future studies

While this study implemented gamified teaching in statistics courses using Classcraft software, the effectiveness and applicability of gamified teaching may vary across different subjects, learning objectives, and student characteristics. As a result, future research can explore the applicability and limitations of gamified teaching in different teaching contexts and propose more principles and examples of gamified teaching design. This study showed that gamified teaching could improve students' learning outcomes and engagement. However, more research is required to find out if these effects endure over time or apply to other learning environments. Future research should investigate whether gamified teaching can enhance students' learning autonomy, strategies, and attitudes, and explore if these improved learning qualities transfer to other subjects or courses. Therefore, future research can conduct longer-term experiments and use more diverse evaluation methods to examine the long-term impact and transfer effects of gamified teaching.

This study mainly explores the benefits and feelings of gamified teaching from the learners' perspective. However, gamified teaching may also involve ethical and social issues, such as whether gamified teaching will affect learners' autonomy, privacy, fairness, etc., and whether gamified teaching will cause competition, exclusion, pressure, etc. Therefore, future research can explore the potential risks and challenges of gamified teaching from an ethical and social perspective and propose corresponding solutions and suggestions.

This study was financially partially supported by the Taiwan Ministry of Education Teaching Practice Research Program (PBM1110361).

CRediT authorship contribution statement

Chih-Chien Wang: Writing – original draft, Supervision, Conceptualization. **Shu-Chen Chang:** Writing – review & editing, Validation, Resources, Formal analysis, Data curation. **Yu Han Yu:** Visualization, Validation, Software.

Declaration of competing interest

This study was financially partially supported by the Taiwan Ministry of Education Teaching Practice Research Program (PBM1110361).

Data availability

Data will be made available on request.

References

[1] I. Bouchrika, N. Harrati, V. Wanick, G. Wills, Exploring the impact of gamification on student engagement and involvement with e-learning systems, *Interact. Learn. Environ.* 29 (8) (2021) 1244–1257.
[2] L.-M. Putz, F. Hofbauer, H. Treiblmaier, Can gamification help to improve education? Findings from a longitudinal study, *Comput. Hum. Behav.* 110 (2020) 106392.
[3] K. Seaborn, D.I. Fels, Gamification in theory and action: a survey, *Int. J. Hum Comput Stud.* 74 (2015) 14–31, https://doi.org/10.1016/j.ijhcs.2014.09.006.
[4] Bellotti, F., Kapralos, B., Lee, K., Moreno-Ger, P., & Berta, R. (2013). Assessment in and of serious games: an overview. *Advances in human-computer interaction, 2013*.
[5] P. Buckley, E. Doyle, Gamification and student motivation, *Interact. Learn. Environ.* 24 (6) (2016) 1162–1175.
[6] R.N. Landers, Developing a theory of gamified learning: Linking serious games and gamification of learning, *Simul. Gaming* 45 (6) (2014) 752–768.
[7] E. Novak, T.E. Johnson, G. Tenenbaum, V.J. Shute, Effects of an instructional gaming characteristic on learning effectiveness, efficiency, and engagement: using a storyline for teaching basic statistical skills, *Interact. Learn. Environ.* 24 (3) (2016) 523–538.
[8] M. Sailer, M. Sailer, Gamification of in-class activities in flipped classroom lectures, *Br. J. Educ. Technol.* 52 (1) (2021) 75–90.
[9] J. Chin, R. Dukes, W. Gamson, Assessment in simulation and gaming: a review of the last 40 years, *Simul. Gaming* 40 (4) (2009) 553–568.
[10] J. Hamari, J. Koivisto, H. Sarsa, Does gamification work?–a literature review of empirical studies on gamification. 2014 47th Hawaii International Conference on System Sciences, 2014.
[11] T. Jagušt, I. Botički, H.-J. So, Examining competitive, collaborative and adaptive gamification in young learners' math learning, *Comput. Educ.* 125 (2018) 444–457.
[12] J. Koivisto, J. Hamari, Demographic differences in perceived benefits from gamification, *Comput. Hum. Behav.* 35 (2014) 179–188.
[13] M.E. Sousa-Vieira, J.C. López-Ardao, M. Fernández-Veiga, M. Rodríguez-Pérez, C. López-García, Mining relationships in learning-oriented social networks, *Comput. Appl. Eng. Educ.* 25 (5) (2017) 769–784.
[14] P.-S.-D. Chen, A.D. Lambert, K.R. Guidry, Engaging online learners: the impact of Web-based learning technology on college student engagement, *Comput. Educ.* 54 (4) (2010) 1222–1232.

[15] M.B. Armstrong, R.N. Landers, Gamification of employee training and development, *Int. J. Train. Dev.* 22 (2) (2018) 162–169.
[16] H. Dehghanzadeh, H. Fardanesh, J. Hatami, E. Talaee, O. Noroozi, Using gamification to support learning English as a second language: a systematic review, *Comput. Assist. Lang. Learn.* 34 (7) (2021) 934–957, https://doi.org/10.1080/09588221.2019.1648298.
[17] H. Dehghanzadeh, M. Farrokhnia, H. Dehghanzadeh, K. Taghipour, O. Noroozi, Using gamification to support learning in K-12 education: a systematic literature review, *Br. J. Educ. Technol.*. (2023).
[18] P. Denny, F. McDonald, R. Empson, P. Kelly, A. Petersen, Empirical support for a causal relationship between gamification and learning outcomes. Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, 2018.
[19] E. Avedon, The study of games, Wiley, Philadelphia, PA, 1971.
[20] J. Juul, M. Copier, J. Raessens, Level up: digital games research. *Conference Proceedings*, 2003.
[21] K. Salen, K.S. Tekinbaş, E. Zimmerman, Rules of play: Game design fundamentals, MIT press, 2004.
[22] S. Deterding, D. Dixon, R. Khaled, L. Nacke, From game design elements to gamefulness: defining" gamification". Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments, 2011.
[23] J. Juul, The game, the player, the world: Looking for a heart of gameness, *Plurais Revista Multidisciplinar* 1 (2) (2010).
[24] Y.-F. Wang, Y.-F. Hsu, K. Fang, The key elements of gamification in corporate training – The Delphi method, *Entertain. Comput.* 40 (2022) 100463, https://doi.org/10.1016/j.entcom.2021.100463.
[25] S. Deterding, Gamification: designing for motivation, *Interactions* 19 (4) (2012) 14–17, https://doi.org/10.1145/2212877.2212883.
[26] A. Ahmad, F. Zeshan, M.S. Khan, R. Marriam, A. Ali, A. Samreen, The impact of gamification on learning outcomes of computer science majors, *ACM Transactions on Computing Education (TOCE)* 20 (2) (2020) 1–25.
[27] J.L. Plass, B.D. Homer, C.K. Kinzer, Foundations of game-based learning, *Educ. Psychol.* 50 (4) (2015) 258–283.
[28] Werbach, K., & Hunter, D. (2012). For the win: How game thinking can revolutionize your business.[IBooks]. In.
[29] K. Huotari, J. Hamari, *Defining gamification–a service marketing perspective, ACM J* (2012) 17–22.
[30] S.K. Banihashem, H. Dehghanzadeh, D. Clark, O. Noroozi, H.J.A. Biemans, Learning analytics for online game-Based learning: a systematic literature review, *Behav. Inform. Technol.* 1–28 (2023), https://doi.org/10.1080/0144929X.2023.2255301.
[31] O. Noroozi, The effects of a digital dialogue game on higher education students' argumentation-based learning, *International Journal of Educational and Pedagogical Sciences* 10 (12) (2017) 4062–4065.
[32] O. Noroozi, Considering students' epistemic beliefs to facilitate their argumentative discourse and attitudinal change with a digital dialogue game, *Innov. Educ. Teach. Int.* 55 (3) (2018) 357–365, https://doi.org/10.1080/14703297.2016.1208112.
[33] O. Noroozi, S. McAlister, M. Mulder, Impacts of a digital dialogue game and epistemic beliefs on argumentative discourse and willingness to argue, *International Review of Research in Open and Distributed Learning* 17 (3) (2016) 208–230.
[34] M.E. Bredemeier, C.S. Greenblat, The educational effectiveness of simulation games: a synthesis of findings, *Simulation & Games* 12 (3) (1981) 307–332.
[35] Greenfield, P. M. (1984). A theory of the teacher in the learning activities of everyday life. *Everyday cognition: Its development in social context*.
[36] R. Matovu, J.C. Nwokeji, T. Holmes, T. Rahman, Teaching and learning cybersecurity awareness with gamification in smaller universities and colleges. 2022 IEEE Frontiers in Education Conference (FIE), 2022.
[37] D. Dicheva, C. Dichev, G. Agre, G. Angelova, Gamification in education: a systematic mapping study, *J. Educ. Technol. Soc.* 18 (3) (2015) 75–88.
[38] I. Blohm, J.M. Leimeister, Gamification design of IT-based enhancing services for motivational support and behavioral change, *Bus. Inf. Syst. Eng.* 5 (2013) 275–278, https://doi.org/10.1007/s12599-013-0273-5.
[39] K.M. Kapp, *The gamification of learning and instruction: game-based methods and strategies for training and education*, John Wiley & Sons, 2012.
[40] R. Raymer, Gamification: using game mechanics to enhance eLearning, *Elearn* 2011 (9) (2011).
[41] R.N. Landers, K.N. Bauer, R.C. Callan, Gamification of task performance with leaderboards: a goal setting experiment, *Comput. Hum. Behav.* 71 (2017) 508–515.
[42] R.N. Landers, E.M. Auer, A.B. Helms, S. Marin, M.B. Armstrong, Gamification of adult learning: gamifying employee training and development, *The Cambridge Handbook of Technology and Employee Behavior* (2019) 271–295.
[43] J. Simões, R.D. Redondo, A.F. Vilas, A social gamification framework for a K-6 learning platform, *Comput. Hum. Behav.* 29 (2) (2013) 345–353.
[44] T. Rashid, H.M. Asghar, Technology use, self-directed learning, student engagement and academic performance: examining the interrelations, *Comput. Hum. Behav.* 63 (2016) 604–612.
[45] R.N. Landers, R.C. Callan, Casual social games as serious games: the psychology of gamification in undergraduate education and employee training, in: *Serious Games and Edutainment Applications*, Springer, 2011, pp. 399–423.
[46] B. Reeves, J.L. Read, *Total engagement: How games and virtual worlds are changing the way people work and businesses compete*, Harvard Business Press, 2009.
[47] L.A. Annetta, J. Minogue, S.Y. Holmes, M.-T. Cheng, Investigating the impact of video games on high school students' engagement and learning about genetics, *Comput. Educ.* 53 (1) (2009) 74–85.
[48] R. Junco, Too much face and not enough books: the relationship between multiple indices of Facebook use and academic performance, *Comput. Hum. Behav.* 28 (1) (2012) 187–198.
[49] R. Junco, G. Heiberger, E. Loken, The effect of Twitter on college student engagement and grades, *J. Comput. Assist. Learn.* 27 (2) (2011) 119–132.
[50] M. Patera, S. Draper, M. Naef, Exploring magic cottage: a virtual reality environment for stimulating children's imaginative writing, *Interact. Learn. Environ.* 16 (3) (2008) 245–263.
[51] C.B. Fried, In-class laptop use and its effects on student learning, *Comput. Educ.* 50 (3) (2008) 906–914.
[52] S. Juniu, Use of technology for constructivist learning in a performance assessment class, *Meas. Phys. Educ. Exerc. Sci.* 10 (1) (2006) 67–79.
[53] C. Rust, B. O'Donovan, M. Price, A social constructivist assessment process model: how the research literature shows us this could be best practice, *Assess. Eval. High. Educ.* 30 (3) (2005) 231–240.
[54] B.E. Weaver, L.B. Nilson, Laptops in class: What are they good for? What can you do with them? *New Dir. Teach. Learn.* 2005 (101) (2005) 3–13.
[55] D.H. White, L. Robertson, Implementing assistive technologies: a study on co-learning in the Canadian elementary school context, *Comput. Hum. Behav.* 51 (2015) 1268–1275.
[56] R. Williams, R. Karousou, J. Mackness, Emergent learning and learning ecologies in Web 2.0, *International Review of Research in Open and Distributed Learning* 12 (3) (2011) 39–59.
[57] T.F.N. Laird, G.D. Kuh, Student experiences with information technology and their relationship to other aspects of student engagement, *Res. High. Educ.* 46 (2) (2005) 211–233.
[58] V. Mehdinezhad, First year students' engagement at the university, *International Online Journal of Educational Sciences* 3 (1) (2011).
[59] R.M. Carini, G.D. Kuh, S.P. Klein, Student engagement and student learning: testing the linkages, *Res. High. Educ.* 47 (1) (2006) 1–32.
[60] P. Ewell, An analysis of relationships between NSSE and selected student learning outcomes measures for seniors attending public institutions in South Dakota, National Center for Higher Education Management Systems, Boulder, CO, 2002.
[61] S.P. Klein, G. Kuh, M. Chun, L. Hamilton, R. Shavelson, An approach to measuring cognitive outcomes across higher education institutions, *Res. High. Educ.* 46 (3) (2005) 251–276.
[62] Pascarella, E. T., & Terenzini, P. T. (2005). *How College Affects Students: A Third Decade of Research. Volume 2*. ERIC.
[63] G.D. Kuh, What we're learning about student engagement from NSSE: Benchmarks for effective educational practices, *Change: the Magazine of Higher Learning* 35 (2003) 24–32.
[64] P. Cheng, R. Ding, The effect of online review exercises on student course engagement and learning performance: a case study of an introductory financial accounting course at an international joint venture university, *J. Account. Educ.* 54 (2021) 100699.
[65] H.A. Mennenga, Student engagement and examination performance in a team-based learning course, *J. Nurs. Educ.* 52 (8) (2013) 475–479.
[66] A. Moubayed, M. Injadat, A. Shami, H. Lutfiyya, Relationship between student engagement and performance in e-learning environment using association rules. 2018 IEEE World Engineering Education Conference (EDUNINE), 2018.
[67] L.S. Shulman, Making differences: a table of learning, *Change: the Magazine of Higher Learning* 34 (2002) 36–44.
[68] HMH. (2024). *Classcraft Studios Inc.* Retrieved Sept. 3 from https://marketplace.classcraft.com/en.
[69] M.M. Handelsman, W.L. Briggs, N. Sullivan, A. Towler, A measure of college student course engagement, *J. Educ. Res.* 98 (3) (2005) 184–192.
[70] J.F. Hair, W.C. Black, B.J. Babin, R.E. Anderson, R.L. Tatham, Multivariate data analysis, Pearson Prentice Hall Upper Saddle River, 2006. In: NJ.